resents a row and column. Each pixel contains information on light intensity level measured in grayscale for black and white, or red, green, and blue values, or a combination of the three colours. This is the reason why the lighting system and lens play a vital role in defining the quality of a pixel or the amount of information in a pixel.

## Image sensor

The image sensor is the heart of the camera. It is important to ensure that camera has the right sensor resolution for your application.

The image sensor is made up of a large number of squares called photo sites. Each photo site corresponds to pixels in an image. So, if you say 5MP pixel camera, it means it produces an image using 5,000,000 pixels, which in turn will have 5,000,000 photo sites in its image sensor. In practical applications, if an industrial camera specifies its frame size as 2560 × 2048 pixels, it means a pixel density of 5,242,880. In short, it means five megapixels.

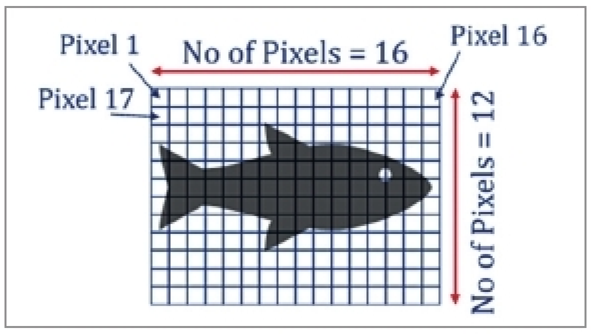


Fig. 5: A picture is a matrix of pixels

*Image sensor properties.* Fig. 8 shows how pixel information is converted into an image. Light energy is assumed as illumination reflected from the pixel, that is, photon energy captured by the camera lens. The individual pixel information is absorbed by the respective photocells in the sensor. The camera sensor converts incoming photon energy into electrical charges, which are then further processed to generate images.

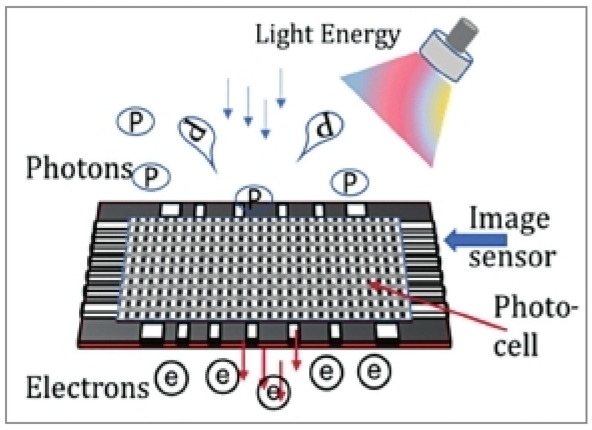


Fig. 8: The camera sensor converts incoming photon energy into electrical charges

The number of electrons you get for the number of photons is called quantum efficiency.

- The higher the quantum efficiency, the more is the information available in an image.
- The size of the pixel is proportional to the quality of light collected.
- The bigger the sensor, the higher the image quality. The number of pixels besides their size also plays an important role.

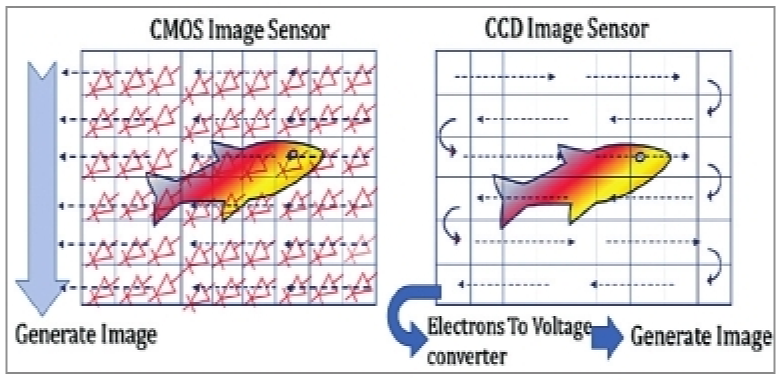


Fig. 9: Types of sensors

## Types of sensors

A charge-coupled device (CCD) is an integrated circuit containing an array of linked or coupled capacitors. Under the control of an external circuit, each capacitor can transfer its electric charge to a neighbouring capacitor. CCD sensors are a major technology used in digital imaging. In a CCD image sensor, pixels are represented by p-doped metal-oxide-semiconductor (MOS) capacitors. These MOS capacitors, the basic building blocks of a CCD, are biased above the threshold for inversion when image acquisition begins, allowing the conversion of incoming photons into electron charges at the semiconductor-oxide interface

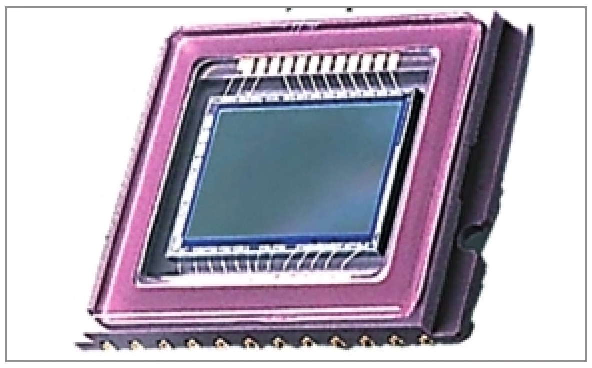

Fig. 10: Fuji Super CCD

Complementary metal oxide semiconductor (CMOS) uses complementary and symmetrical pairs of p-type and n-type MOSFETs for logic functions. CMOS technology is used for constructing integrated circuit (IC) chips, including microprocessors, microcontrollers, memory chips (including CMOS BIOS), and other digital logic circuits. It has replaced the earlier transistor-transistor logic (TTL) technology. It is powerful, sensitive, fast, and affordable. In this, electron to voltage conversion happens at individual pixel itself because an

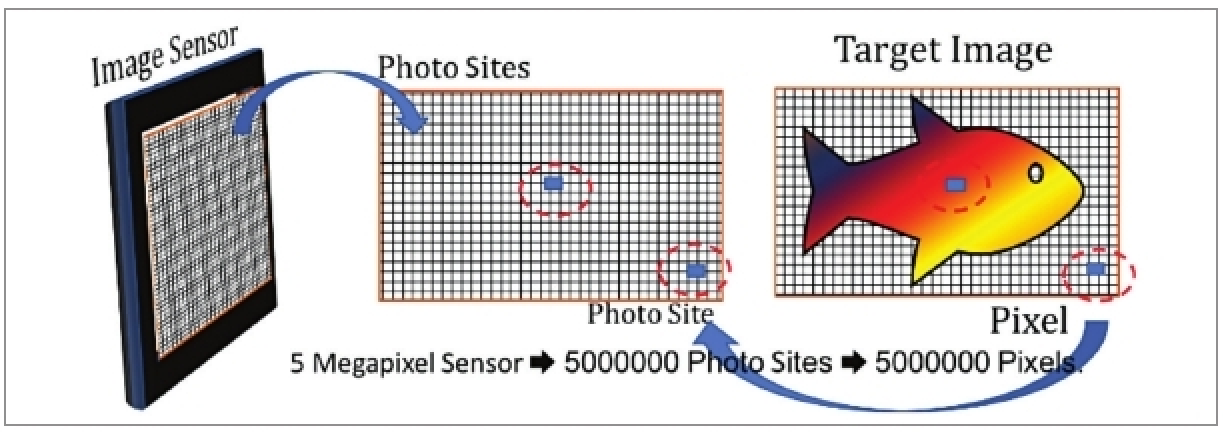


Fig. 6: Image sensor is an important part of a camera for taking high-resolution pictures

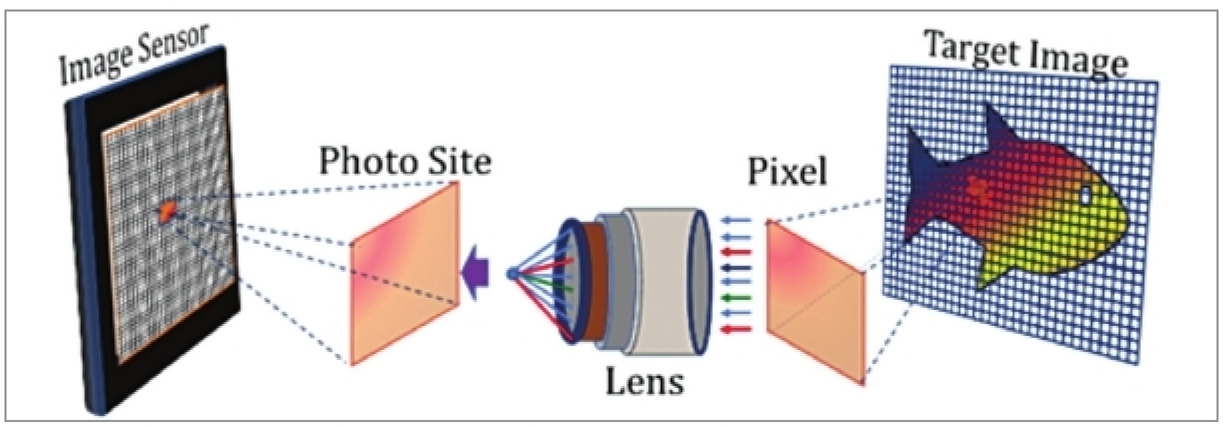


Fig. 7: Image sensor comprises a large number of squares called photo sites